

AdaTS: Adaptive Token Sampling for Efficient Speech Language Models

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Abstract

Speech Language Models (SLMs) have demonstrated strong speech understanding capabilities by incorporating speech tokens into large LMs (LLMs). However, most designs are i) token-intensive, as a large part of the LLM context is allotted to speech tokens, and ii) inefficient, as speech tokens are often temporally redundant, hindering SLMs' capabilities to handle long-form tasks. To address both issues, we propose AdaTS, a dynamic sampling method that adaptively merges tokens where the speech signal is less informative. Our approach reduces the length of the speech signal seen by the LLM by $2\times$ or more, with performance often better than standard downsampling across automatic speech recognition (ASR), speech question-answering (SQA), and speech translation (ST). Through an extensive empirical analysis, we show how SLMs using AdaTS retain strong speech understanding capabilities, while substantially reducing token redundancy and inference cost by 40%.¹

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction

Recent advances in large language models (LLMs) have enabled the extension of their general capabilities to other modalities beyond text [1, 2, 3]. Among these efforts, speech language models (SLMs) have emerged as a promising direction, aiming to equip LLMs with the ability to process, understand, and generate speech signals directly, rather than relying solely on text-based representations [4, 5, 6]. Despite this progress, current SLMs are severely bottlenecked by long context lengths, due to the quadratic complexity of self-attention mechanisms in transformers [7]. More so than text, speech inputs are *token intensive*—for instance, modern speech encoders, sampling every 20ms (50 Hz) or finer, produce thousands of tokens per minute of audio [8, 9]. As a result, even a 10-minute clip may exceed 30K tokens, which is at the upper limit of the effective context length for many LLMs [10], making real-world scenarios with long-form audio entirely impractical. Moreover, speech tokens are largely *redundant*, with less than 50% of tokens actually being attended to by the models [11, 7].

Token *sampling* has been proposed to overcome the challenges imposed by the longer contexts of multimodal tokens. This approach is not only highly effective but also practical to implement as both a test-time technique or as part of the training framework. The use of token pruning-based strategies has been widely studied for both speech and text modalities, where tokens with low importance, usually determined via the cumulative attention scores [12, 13], are discarded. For sequence-to-sequence

tasks, subsequent research has instead focused on subsampling from the input sequence through temporal pooling or convolutional downsampling [14, 1, 15]. While they can reduce overall sequence length, these methods are *static* and *indiscriminate*, treating all tokens uniformly regardless of their content. This creates a fundamental trade-off between the compression rate and semantic preservation, often squeezing content-critical regions together with low-information segments which can be sub-optimal [16].

In this work, we propose ADATS, a dynamic sampling framework for efficient SLMs that adapts to each speech signal. ADATS operates directly on the output of a pretrained speech encoder, following which we apply a *score-and-merge* mechanism based on pairwise token similarity. Unlike uniform downsampling, speech tokens are adaptively split into local subgroups and merged according to their information density, producing coarse representations in redundant regions while preserving high-resolution detail in content-bearing segments. We build upon the success of small-scale LLMs [17, 18, 19] and train ADATS with a standard two-stage SLM training pipeline. We demonstrate through evaluations across several tasks that ADATS preserves temporal consistency and fine-grained phonetic content while reducing overall redundancy. On automatic speech recognition (ASR), speech translation (ST), and speech question answering (SQA), including long-form SQA, we achieve **up to $4\times$ compression and $2\times$ on average** while consistently improving upon downstream performance. In addition, ADATS reduces inference cost by **approximately 40%** compared to convolutional downsampling at smaller model scales for better performance. Our findings open the pathways of efficient SLM by showing that scaling is not only about larger backbone LLMs or longer contexts, but also about how speech tokens are utilised.

2. Adaptive Speech Sampling

ADATS follows a two-stage training strategy where we first align the output space of the speech encoder and the input space of the text decoder keeping the components frozen, then we learn a full fine-tuning stage where the decoder is trainable.

2.1. Sampling Module

From a pre-trained speech encoder, we first generate a sequence of high-dimensional feature vectors $A = a_1, a_2, \dots, a_n$. Next, we compute pair-wise token similarity and merge a consecutive subgroup of tokens with $\text{sim}(a_i, a_j) > t$ for a chosen threshold t , which are merged after all similarities are computed as follows:

^{**}indicates the corresponding author.

¹Code: <https://github.com/sonalsannigrahi/AdaTS>

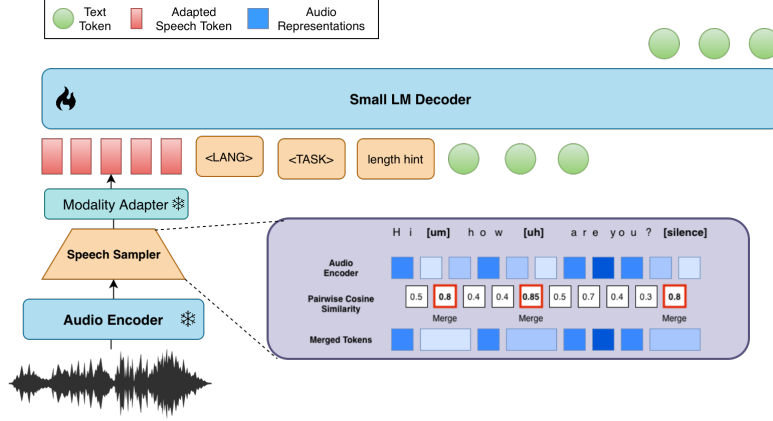


Figure 1: Illustration of ADATS including the sampling module which consists of three steps: obtaining representations from the audio encoder, computing pairwise cosine similarity, and merging tokens.

$$\text{sim}(a_i, a_j) = \frac{a_i a_j}{|a_i| |a_j|}, a_{\text{merged}} = \frac{\sum_{i=1}^N (1 - \text{sim}(a_i, a_{i-1})) a_i}{\sum_{i=1}^N (1 - \text{sim}(a_i, a_{i-1}))}, \quad (1)$$

where $0 < i$ and $N < n$ is the total number of tokens² with similarity below t . We then replace tokens $a_i, \dots, a_N$ with a_{merged} for each subgroup in the feature vector A to produce the output sequence A' resulting in fewer tokens overall. a_{merged} is the normalised, weighted sum of each element in feature vector where the weight is the inverse of the similarity. Conceptually, this leads to higher weights for more *dissimilar* units and lower for others. For our choice of speech encoder, which is trained with contrastive cosine loss, to produce A' , cosine similarity is an appropriate choice to measure token similarity [20, 21].

2.2. Model Architecture

The three primary components are the speech encoder, projector, and the LLM decoder. In line with recent multimodal encoder-decoder models [14, 22, 23], we follow a two-stage training approach. First, we learn a linear projector between the frozen speech encoder and frozen LLM decoder. Then, in the second stage, we unfreeze the LLM and fully finetune the model with a task-specific data mix. In the first training stage, we align the embedding spaces of the speech encoder and text decoder with linear multi-layer perceptron (MLP) layers. Our speech encoder receives 80-dimensional Mel-filterbank frame representations and transforms them into multidimensional embeddings. These outputs are then processed by a **modality and length adapter** to project them to the input embedding space of the decoder. During this stage, all components are kept frozen and only the pre-encoder layers are trainable using ASR data. We follow all hyperparameters suggested in [14]. Following the modality alignment, we concatenate the audio embeddings to the text input embeddings and perform instruction tuning (IT) using a suite of speech-to-text tasks for English. We train on ASR in English, SQA in English, and ST from English to 10 other languages.³

²We consider the feature vectors to be *token representations* and use it interchangeably

³French, German, Italian, Dutch, Korean, Portuguese, Russian, Chinese, and Spanish.

2.3. Training Setup

We train the model using a standard cross-entropy loss with a cosine scheduler on reference transcripts on 4 H100 GPUs for 4 days. For both stages, we use 20 warmup steps with 128 gradient accumulation steps. We use AdamW [24] as our optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.95$. We also use bfloat16 mixed precision. We set our learning rate as 6×10^{-6} for the encoder, 2×10^{-5} for the decoder, and 2×10^{-4} for the adapter.

2.4. Data

We utilise a mix of natural and synthetic data sources covering ASR, SQA, and ST to complement existing data for under-resourced tasks and language pairs. We use CC-BY licensed data to promote open science and reproducibility. In total, we collect 80K hours of data for all three tasks.

ASR: We use LibriSpeech (LS) [25], VoxPopuli [26], CommonVoice 16.1 [27] during the modality alignment phase. In the IT phase, we use all previous data and add Peoples Speech [28], Giga Speech [29], and the English split of MLS [30].

ST: We use gold-standard CoVoST-2 [31] for En-De and a pseudo-labelled speech translation corpora, Spite⁴, to augment our ST data mix for remaining language pairs. We use the version of Spite translated using TowerPlus-9B and with transcripts from CommonVoice 16.1 as the translation quality of TowerPlus-9B is the highest among the models released [32].

SQA: We use the SpokenSQuAD [33], SLUE SQA-5 [34], and LibriSQA [35] datasets which contain a mix of synthesized speech as well as natural human speech sources paired with text targets. We clean the QA pairs obtained from SpokenSQuAD and SLUE SQA-5 using Qwen2.5 7B to reformulate the text-based answers into stand alone sentences [36].

3. Experiments

We train and evaluate ADATS on three tasks: ASR, ST, SQA, and long-form SQA. We first discuss the decoding method used, followed by different system settings. Next, we detail comparable baseline approaches. Lastly, we list all the evaluation datasets and metrics used for evaluation.

⁴<https://huggingface.co/datasets/bpop/spite-CV16-TP9B>

Table 1: Main results on ASR, SQA, and ST. We report baselines above and highlight models with ADATS.

Models	ASR (↓)				SQA (↑)		ST (↑)	
	Clean	Other	VoxPopuli	FLEURS	Spoken SQuAD	SLUE SQA	LongSpeechEval	FLEURS
Pooling	5.3	8.2	9.8	11.2	45.1	27.3	2.7	80.1
WLQF	3.6	4.5	10.2	10.1	53.1	30.4	3.0	81.3
FLS-1.5B	5.2	7.8	10.1	12.1	50.4	32.3	3.2	80.2
FLS-7B	4.1	7.2	9.8	10.3	-	-	3.6	-
LLaMA 3.2 1B	4.3	10.2	15.5	12.3	53.5	32.6	3.2	81.1
EuroLLM 1.7B	5.6	9.8	14.4	10.9	52.3	31.4	3.1	78.6
Qwen 2.5 1.5B	2.9	6.2	9.5	8.5	58.9	37.5	3.5	82.0

3.1. Evaluation Set-up

We decode with zero shot prompting using task-specific tags unless specified otherwise. For all tasks, we generate using beam search with beam size 3, a repetition penalty of 1.6, up to 1024 tokens. We constrain the text generation using a target language and task token. The task tag is limited to `<|transcribe|>`, `<|translate|>`, or `<|reply|>` for ASR, ST, SQA respectively. We use the relevant ISO codes for each language tag.

3.2. System Settings

We fix the speech encoder in ADATS as Wav2Vec2Bert and vary the language decoder. We restrict our experiments to small-scale language models and use the base variants of Qwen2.5 1.5B, Llama 3.2 1B, and EuroLLM 1.7B. Within the sampling module, we report results with the following variations:

- **Threshold:** We modulate the similarity threshold between 0.7 and 0.9 in 0.05 increments.
- **Pooling method:** We report three approaches: 1) average uniformly across the entire token subgroup obtained for a certain threshold t , 2) weight each token in the subgroup by $(1 - s)$ where s is the previously computed pairwise cosine similarity per (1), 3) prune all but the first token in the subgroup (top-1).
- **Training stage:** Within the two-stage training set-up, we include the sampling module during the MA phase, the IT phase, or in both.

3.3. Baselines

We consider two approaches: static downsampling (conv-based and WLQF) and content-dependent sampling (CTC-based) [37]. We fix the speech encoder to Wav2Vec2Bert and use Qwen2.5 1.5B as the backbone decoder for all cases.

- **Conv-based:** We use 3 1D-convolutional layers with stride 2 and kernel size 3 after the modality adapter.
- **WLQF:** We use a Window-level Q-former (WLQF) [6] to replace the modality and length adapter. The module takes in variable-length outputs from the speech encoder and splits them into non-overlapping windows of size N which is then passed to a Q-former [38]. We set $N = 0.33$ seconds and the number of queries to 1 as per [6]. We set the adapter kernel size to 8 and stride to 8 per the best setting in [37].
- **CTC-based:** FastLongSpeech (FLS) [39] uses an iterative fusion strategy trained with a CTC decoder to merge speech tokens. FLS is trained with Qwen2 Audio 7B as the base LLM, to be comparable we report results with the 7B and 1.5B variant.

3.4. Datasets

We report Word Error Rate (WER) on the LS test split [25], VoxPopuli [26], and FLEURS [40] for ASR. We report average COMET-22 [41] on en→xx directions from FLEURS ST for ST across all language pairs considered. For SQA, we report F1 accuracy on the test split of Spoken SQuAD [33] and frame-F1, defined as F1 accuracy on the answer spans, on SLUE SQA-5 [35]. We further report results on LongSpeechEval (LS Eval) ⁵, where the average duration is 2 minutes, using a LLM-as-a-judge protocol as described in [39].

4. Main Results and Discussion

Table 1 presents our main results on ASR, SQA, and ST. We use ASR as a proxy for speech content preservation and SQA and ST to evaluate overall speech understanding. Unless mentioned otherwise, we refer to ADATS with Qwen2.5 1.5B as the decoder.

4.1. Speech Content Preservation

We observe in Table 1, that across all ASR datasets, ADATS consistently outperforms other methods. Considering both static downsampling as well as content-dependent sampling, we find that ADATS remains the least sensitive to the subsampling of the speech sequence and scale of the LLMs used. This finding suggests that we are able to effectively preserve fine-grained details required to transcribe content accurately while reducing the overall redundancy of the input. We discuss exact compression rates achieved in Section 4.5. Among content-dependent methods, the FastLongSpeech (FLS) variants trained with 1.5B and 7B backbone LLMs exhibit a clear performance gap, showing that the size of the decoder is crucial to final performance for this approach—a dependency ADATS does not share, as it achieves competitive results even with smaller backbones. Considering variants with several small-scale backbone LMs, we find that the choice of backbone strongly determines overall performance, with Qwen 2.5 1.5B emerging as a consistently strong foundation that yields the best results across ASR and ST tasks alike. However, despite the same backbone, ADATS improves upon uniform pooling, even surpasses FLS-7B, demonstrating that effective compression can compensate for reduced model scale.

4.2. Speech Understanding

In Table 1, for both tasks of short-form SQA and AST, ADATS outperforms other approaches. Notably, for SQA, pooling has

⁵<https://huggingface.co/datasets/ICTNLP/LongSpeech-Eval>

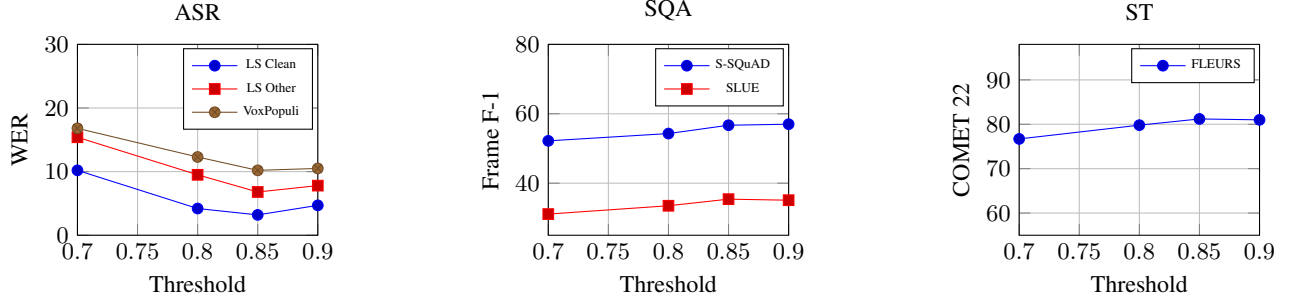


Figure 2: Performance vs. Threshold for ASR, SQA, and ST.

Table 2: Merge Strategies on LS Clean, SLUE, and FLEURS.

Strategy	ASR ($\downarrow$)	SQA ($\uparrow$)	ST ($\uparrow$)
Avg	3.2	35.4	81.2
Weighted Avg	2.9	37.5	82.0
Top-1	3.0	37.2	81.0

Table 3: Training Stage Ablations. Results on LS Clean, SLUE, and FLEURS.

MA	IT	ASR ($\downarrow$)	SQA ($\uparrow$)	ST ($\uparrow$)
$\times$	$\times$	5.3	25.6	80.1
$\times$	$\checkmark$	3.2	35.4	81.2
$\checkmark$	$\times$	20.3	20.4	58.8
$\checkmark$	$\checkmark$	3.6	29.7	80.3

Table 4: Compression Ratio Across Test Tasks with $t = 0.85$.

Dataset	ASR	SQA	ST
Min	1.23	1.16	1.25
Max	4.16	3.21	4.24
Avg	1.76	1.65	1.85

the largest performance gap with other approaches showing the limitations of static downsampling. WLQF improves considerably upon simple convolutional downsampling and achieves comparable results to CTC-based FLS. Compared to ASR, FLS is a stronger model for SQA tasks showing that CTC-based decoding is able to better retain the overall information in the speech input but is worse at preserving the required phonetic characteristics for accurate ASR. ADATS is able to improve upon the task-specific performance of FLS as well the overall phonetic content preservation of WLQF. On long-form QA, LSEval, we find that ADATS is able to perform competitively with FLS-7B despite training on average with audios shorter than 30s.

4.3. Threshold Sensitivity

Figure 2 illustrates the performance of our evaluation suite against different threshold values t . Across all tasks, we observe a similar trend with respect to t , where lower values degrade performance. Comparing the three tasks, we note that the ASR and ST are much more sensitive to t with significant performance drops with changes in the value. However, in the SQA task the effect of the threshold is much less pronounced. This suggests that tasks which require high content preservation also in turn require a more fine grained choice of t , whereas tasks requiring general speech understanding are more robust to larger downsampling. Fixing $t = 0.85$, we now investigate the optimal merge strategy. Table 2 reports results with three pooling methods: averaging, averaging using weights as expressed in Eq. 1, and pruning all tokens but the first in the token subgroup (Top-1). While simple averaging provides a strong baseline, we are able to further improve upon it by weighted average which we find to be the best performing approach. Pruning as a strategy is surprisingly effective as well in downstream tasks suggesting that the token subgroups obtained by sampling are indeed low in information density.

4.4. Training Stages

Table 3 shows our results on activating the sampling module in MA, IT, or both. Across all tasks, the best strategy is to

perform MA with the full sequences and then perform IT with the subsampled tokens. This is in line with previous literature where the sampling strategy is often applied during the IT phase [7]. However, applying compression only in MA leads to a subpar model that is unable to perform speech-text tasks. Therefore, the alignment learnt during MA is crucial for the downstream performance of the model. We hypothesise that the redundancy of similar speech tokens is required in MA as they still bring a useful speech signal, despite being uniform, which is beneficial in terms of the **quantity**. This is lost if the signal is heavily compressed at this stage by subsampling.

4.5. Efficiency

Table 4 reports the compression rates achieved by ADATS, averaging across datasets in each evaluation task. We report compression statistics at $t = 0.85$, which we found work best among the tasks considered. On average, we obtained a 2x compression rate across all tasks. In particular, for ASR and ST tasks, we are able to compress more aggressively for the same threshold when applied to SQA showing that the actual compression rate achieved is inherently dataset dependent. Lastly, ADATS requires 2.14 TFLOPS for inference⁶ on a 10s audio while convolutional downsampling requires 3.23 TFLOPs, yielding a 34% reduction in compute. For a comparable performance, FLS-7B requires 8.54 TFLOPs representing 4 times the computational effort.

5. Conclusion

We presented ADATS, a parameter-free sampling method based on pairwise token similarity. Our results show that ADATS achieves competitive performance on several tasks across benchmarks while reducing the token footprint up to 2x on average and obtaining significant gains in terms on inference efficiency. Lastly, we report the best settings to use similarity-based sampling to guide research in token-based compression strategies.

⁶We computed FLOPS using <https://github.com/MrYxJ/calculate-flops.pytorch>

6. Generative AI Disclosure

No Generative AI tools were used to write any part of this document. AI coding assistants in Github Copilot and Claude Sonnet were used for performing parts of the experiments reported in this manuscript.

7. References

- [1] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang *et al.*, “Qwen2. 5-omni technical report,” *arXiv preprint arXiv:2503.20215*, 2025.
- [2] Q. Fang, S. Guo, Y. Zhou, Z. Ma, S. Zhang, and Y. Feng, “Llama-omni: Seamless speech interaction with large language models,” *arXiv preprint arXiv:2409.06666*, 2024.
- [3] S. Dash, Y. Nan, J. Dang, A. Ahmadian, S. Singh, M. Smith, B. Venkitesh, V. Shmyhlo, V. Aryabumi, W. Beller-Morales *et al.*, “Aya vision: Advancing the frontier of multilingual multimodality,” *arXiv preprint arXiv:2505.08751*, 2025.
- [4] X. Shi, X. Wang, Z. Guo, Y. Wang, P. Zhang, X. Zhang, Z. Guo, H. Hao, Y. Xi, B. Yang *et al.*, “Qwen3-asr technical report,” *arXiv preprint arXiv:2601.21337*, 2026.
- [5] W. Cui, D. Yu, X. Jiao, Z. Meng, G. Zhang, Q. Wang, S. Y. Guo, and I. King, “Recent advances in speech language models: A survey,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 13 943–13 970.
- [6] C. Tang, W. Yu, G. Sun, X. Chen, T. Tan, W. Li, L. Lu, Z. Ma, and C. Zhang, “Salmonn: Towards generic hearing abilities for large language models,” in *The Twelfth International Conference on Learning Representations*, 2024.
- [7] K. Shao, K. Tao, K. Zhang, S. Feng, M. Cai, Y. Shang, H. You, C. Qin, Y. Sui, and H. Wang, “When tokens talk too much: A survey of multimodal long-context token compression across images, videos, and audios,” *arXiv preprint arXiv:2507.20198*, 2025.
- [8] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, “Hubert: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 3451–3460, 2021.
- [9] S. Hu, L. Zhou, S. Liu, S. Chen, L. Meng, H. Hao, J. Pan, X. Liu, J. Li, S. Sivasankaran *et al.*, “Wavlm: Towards robust and adaptive speech large language model,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024, pp. 4552–4572.
- [10] C.-P. Hsieh, S. Sun, S. Kriman, S. Acharya, D. Rekesh, F. Jia, and B. Ginsburg, “Ruler: What’s the real context size of your long-context language models?” in *First Conference on Language Modeling*, 2024.
- [11] K. Shao, K. Tao, C. Qin, H. You, Y. Sui, and H. Wang, “Holitom: Holistic token merging for fast video large language models,” in *NeurIPS*, 2025.
- [12] Y. Shang, M. Cai, B. Xu, Y. J. Lee, and Y. Yan, “Llava-prumerge: Adaptive token reduction for efficient large multimodal models,” *ICCV*, 2025.
- [13] Y. Lin, Y. Fu, J. Zhang, Y. Liu, J. Zhang, J. Sun, H. H. Li, and Y. Chen, “Speechprune: Context-aware token pruning for speech information retrieval,” in *2025 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2025, pp. 1–6.
- [14] G. Attanasio, S. Sannigrahi, B. Peters, and A. Filipe Torres Martins, “Instituto de telecomunicações at IWSLT 2025: Aligning small-scale speech and language models for speech-to-text learning,” in *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, E. Salesky, M. Federico, and A. Anastasopoulos, Eds. Vienna, Austria (in-person and online): Association for Computational Linguistics, Jul. 2025, pp. 347–353. [Online]. Available: <https://aclanthology.org/2025.iwslt-1.36/>
- [15] C. Umberto, L. Xubo, M. Pingchuan, P. Stavros, and P. Maja, “Omni-avsr: Towards unified multimodal speech recognition with large language models,” *arxiv 2511.07253*, 2025.
- [16] H.-J. Chang, S.-w. Yang, and H.-y. Lee, “Distilhubert: Speech representation learning by layer-wise distillation of hidden-unit bert,” in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 7087–7091.
- [17] Microsoft, :, A. Abouelenin, A. Ashfaq, A. Atkinson, H. Awadalla, N. Bach, J. Bao, A. Benhaim, M. Cai, V. Chaudhary, C. Chen, D. Chen, D. Chen, J. Chen, W. Chen, Y.-C. Chen, Y. ling Chen, Q. Dai, X. Dai, R. Fan, M. Gao, M. Gao, A. Garg, A. Goswami, J. Hao, A. Hendy, Y. Hu, X. Jin, M. Khademi, D. Kim, Y. J. Kim, G. Lee, J. Li, Y. Li, C. Liang, X. Lin, Z. Lin, M. Liu, Y. Liu, G. Lopez, C. Luo, P. Madan, V. Mazalov, A. Mitra, A. Mousavi, A. Nguyen, J. Pan, D. Perez-Becker, J. Platin, T. Portet, K. Qiu, B. Ren, L. Ren, S. Roy, N. Shang, Y. Shen, S. Singhal, S. Som, X. Song, T. Sych, P. Vaddamanu, S. Wang, Y. Wang, Z. Wang, H. Wu, H. Xu, W. Xu, Y. Yang, Z. Yang, D. Yu, I. Zabir, J. Zhang, L. L. Zhang, Y. Zhang, and X. Zhou, “Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.01743>
- [18] J. Xu, Z. Guo, H. Hu, Y. Chu, X. Wang, J. He, Y. Wang, X. Shi, T. He, X. Zhu, Y. Lv, Y. Wang, D. Guo, H. Wang, L. Ma, P. Zhang, X. Zhang, H. Hao, Z. Guo, B. Yang, B. Zhang, Z. Ma, X. Wei, S. Bai, K. Chen, X. Liu, P. Wang, M. Yang, D. Liu, X. Ren, B. Zheng, R. Men, F. Zhou, B. Yu, J. Yang, L. Yu, J. Zhou, and J. Lin, “Qwen3-omni technical report,” 2025. [Online]. Available: <https://arxiv.org/abs/2509.17765>
- [19] P. H. Martins, J. Alves, P. Fernandes, N. M. Guerreiro, R. Rei, A. Farajian, M. Klimaszewski, D. M. Alves, J. Pombal, N. Boizard *et al.*, “Eurollm-9b: Technical report,” *arXiv preprint arXiv:2506.04079*, 2025.
- [20] H. Zhang, J. Zhang, X. Ji, Q. Wang, and F. Zhang, “Dyntok: Dynamic compression of visual tokens for efficient and effective video understanding,” *arXiv preprint arXiv:2506.03990*, 2025.
- [21] K. Zhou, K. Ethayarajh, D. Card, and D. Jurafsky, “Problems with cosine as a measure of embedding similarity for high frequency words,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 401–423. [Online]. Available: <https://aclanthology.org/2022.acl-short.45/>
- [22] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang *et al.*, “Qwen2. 5-vl technical report,” *arXiv preprint arXiv:2502.13923*, 2025.
- [23] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan *et al.*, “The llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
- [24] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *ICLR (Poster)*, Y. Bengio and Y. LeCun, Eds., 2015. [Online]. Available: <http://dblp.uni-trier.de/db/conf/iclr/iclr2015.html#KingmaB14>
- [25] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “Librispeech: an asr corpus based on public domain audio books,” in *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.
- [26] C. Wang, M. Riviere, A. Lee, A. Wu, C. Talnikar, D. Haziza, M. Williamson, J. Pino, and E. Dupoux, “Voxpopuli: A large-scale multilingual speech corpus for representation learning, semi-supervised learning and interpretation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 2021, pp. 993–1003.
- [27] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, and G. Weber, “Common voice: A massively-multilingual speech corpus,” in *Proceedings of the twelfth language resources and evaluation conference*, 2020, pp. 4218–4222.

- [28] D. Galvez, G. Damos, J. M. C. Torres, J. F. Cerón, K. Achorn, A. Gopi, D. Kanter, M. Lam, M. Mazumder, and V. J. Reddi, "The people's speech: A large-scale diverse english speech recognition dataset for commercial usage," in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 1)*, 2022.
- [29] G. W. J. D. W.-Q. Z. C. W. D. S. D. P. J. T. J. Z. M. J. S. K. S. W. S. Z. W. Z. X. L. X. Y. Y. W. Y. W. Z. Y. Z. Y. Guoguo Chen, Shuzhou Chai, "Gigaspeech: An evolving, multi-domain asr corpus with 10,000 hours of transcribed audio," in *Proc. Interspeech 2021*, 2021.
- [30] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, and R. Collobert, "Mls: A large-scale multilingual dataset for speech research," 2020.
- [31] C. Wang, A. Wu, and J. Pino, "Covost 2: A massively multilingual speech-to-text translation corpus," 2020.
- [32] R. Rei, N. M. Guerreiro, J. Pombal, J. Alves, P. Teixeira, A. Farajian, and A. F. Martins, "Tower+: Bridging generality and translation specialization in multilingual llms," *arXiv preprint arXiv:2506.17080*, 2025.
- [33] C.-H. Lee, S.-L. Wu, C.-L. Liu, and H.-y. Lee, "Spoken squad: A study of mitigating the impact of speech recognition errors on listening comprehension," in *Proc. Interspeech 2018*, 2018, pp. 3459–3463.
- [34] S. Shon, S. Arora, C.-J. Lin, A. Pasad, F. Wu, R. Sharma, W.-L. Wu, H.-y. Lee, K. Livescu, and S. Watanabe, "Slue phase-2: A benchmark suite of diverse spoken language understanding tasks," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 8906–8937.
- [35] Z. Zhao, Y. Jiang, H. Liu, Y. Wang, and Y. Wang, "Librisqa: A novel dataset and framework for spoken question answering with large language models," *IEEE Transactions on Artificial Intelligence*, 2024.
- [36] B. Lee, M. Zanon-Boito, L. Besacier, and I. Calapodescu, "Naver labs europe submission to the instruction-following track," in *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, 2025, pp. 186–200.
- [37] F. Verdini, P. Melucci, S. Perna, F. Cariaggi, M. Gaido, S. Papi, S. Mazurek, M. Kasztelnik, L. Bentivogli, S. Bratières *et al.*, "How to connect speech foundation models and large language models? what matters and what does not," *arXiv preprint arXiv:2409.17044*, 2024.
- [38] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *International conference on machine learning*. PMLR, 2023, pp. 19 730–19 742.
- [39] S. Guo, S. Zhang, Q. Fang, Z. Ma, Y. Feng *et al.*, "Fastlongspeech: Enhancing large speech-language models for efficient long-speech processing," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [40] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, "Fleurs: Few-shot learning evaluation of universal representations of speech," in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 798–805.
- [41] R. Rei, C. Stewart, A. C. Farinha, and A. Lavie, "Comet: A neural framework for mt evaluation," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 2685–2702.